Formula sheet for Quiz 7

Note to students: you will be handed a copy with the quiz, so nothing here needs memorizing. It lists more than any one question needs, so deciding which applies is still your job.

Convolution shapes

Each filter spans every input channel, so one filter is $k \times k \times C_{\text{in}}$. The number of filters is what you choose, and it becomes C_{out}.

For one spatial axis, with input size W, kernel k, padding p, stride s:

$$W_{\text{out}} = \left\lfloor \frac{W + 2p - k}{s} \right\rfloor + 1 \qquad \textbf{Conv2d}(C_{\text{in}}, C_{\text{out}}, k) = C_{\text{out}}\,(C_{\text{in}}k^2 + 1)$$

$$\textbf{Linear}(n_{\text{in}}, n_{\text{out}}) = n_{\text{out}}(n_{\text{in}} + 1) \qquad \textbf{MaxPool2d}(k, s):\ \ W_{\text{out}} = \left\lfloor \frac{W - k}{s} \right\rfloor + 1$$

A 1×1 convolution is the $k = 1$ case: it leaves H and W alone and maps C_{in} channels to C_{out}. From a $C \times H \times W$ feature map, **flatten** gives $C \cdot H \cdot W$ values and **global average pooling** gives C.

Costs, for C channels in and out: a dense $k \times k$ convolution is C^2k^2 weights, a depthwise stage is Ck^2 and a pointwise stage is C^2, so the separable pair is a ratio of $1/C + 1/k^2$. A grouped convolution with g groups costs $1/g$ of dense.

Batch normalization

Per channel, over the values being used as the statistics:

$$\hat{x} = \frac{x - \mu}{\sqrt{\sigma^2 + \epsilon}} \qquad y = \gamma\hat{x} + \beta$$

γ and β are learned, two parameters per channel.

Residual and dense connections

$$\textbf{residual:}\ \ y = f(x) + x \qquad \textbf{dense:}\ \ y = [\, f(x)\,;\, x\,]$$

$[\,\cdot\,;\,\cdot\,]$ is concatenation along the channel axis, so C channels and D channels give $C + D$.

Boxes

A box is (x_1, y_1, x_2, y_2) in **pixels**, top-left and bottom-right, which is what `torchvision` detectors return, with area $(x_2 - x_1)(y_2 - y_1)$. A box can equally be a centre with a width and height; either way, four numbers.

Intersection over Union

For boxes A and B, the intersection box runs from $(\max(x_1^A, x_1^B),\ \max(y_1^A, y_1^B))$ to $(\min(x_2^A, x_2^B),\ \min(y_2^A, y_2^B))$:

$$I = \max(x_2 - x_1,\, 0) \cdot \max(y_2 - y_1,\, 0) \qquad \text{IoU} = \frac{I}{\text{area}(A) + \text{area}(B) - I}$$

The union subtracts I because adding the two areas counts the overlap twice. IoU is 0 for boxes that do not touch and 1 for identical boxes.

Notation

N batch size, C channels or classes, H and W spatial size, k kernel size, p padding, s stride, g groups, ϵ a small constant keeping a denominator away from zero.